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CFD simulation study of flow equalisation plate model in single-phase immersion liquid cooling for servers

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ABSTRACT

Immersion liquid cooling is progressively being adopted for server cooling applications within data centers. Uneven coolant distribution in liquid-cooled cabinets may cause large temperature difference between servers under some extreme conditions, prompted this study on a flow equalization plate structure. Firstly, the servers, the cabinet and the flow equalisation plate were modelled using Ansys Icepak software, and then the effects of holes radii, inlet layout configurations, and the number of holes on the flow distribution of the FC-40 coolant and the temperature control across servers was investigated. The results indicate that the smaller radii is beneficial in balancing the flowrate within the holes of the flow equalization plate by increasing the flow resistance, yielding a maximum temperature difference and a maximum temperature standard deviation of 2.1 °C and 0.8 °C at r = 7.5 mm. However, excessively small radii can lead to higher internal pressures and flow resistance. With larger hole radii, temperature control is significantly influenced by the inlet flowrate. As the flowrate increases, ΔT and T_{sd} also increase, reaching up to 3.8 °C and 1.3 °C at r = 12.5 mm, respectively. Larger hole radii result in smaller coolant flowrate, leading to lower convective heat transfer coefficient and higher server temperature. Locating inlets on one side creates large low-velocity vortex areas, disrupting server cooling, while center placement enhances thermal performance. When the number of holes is 18, the hole spacing increases, which leads to an increase in the flowrate in the holes and an increase in the convective heat transfer coefficient, which improves the thermal performance factor ε to 1.04.

Introduction

Data centres, which are primarily responsible for storing and processing business and production processes such as social media, big data, bitcoin, artificial intelligence, etc., are dynamically growing in terms of computing capacity, and playing a crucial role as hubs for coordinating network exchanges between different regions. Due to the need for 24/7 uninterrupted operation, data centres consume a significant amount of electricity. In fact, data centres alone will account for approximately 4.5 % of the total electricity consumption in the world in 2025 [1]. In 2016 data centres power demand was 286 TWh, the figure that will rise to 321 TWh in 2030 if today's technological and behavioural trends remain unchanged [2]. These centres consist of various systems, including the IT system such as servers and switches, the cooling system, the power supply and distribution system, the lighting system, and other facilities. Remarkably, the cooling system, which accounts for 30 % to 50 % of the total electricity consumption [3,4,5], evidently present significant potential for energy conservation. Additionally, temperature plays a crucial role in the functioning of electronic equipment. When a local hotspot occurs within the device, it initially impacts the server's operating speed. Additionally, it generates thermal stresses and strains, which may lead to device damage and failure. Consequently, it is essential to explore approaches that both reduce cooling energy consumption and enhance cooling capacity.

Immersion liquid cooling (ILC) has emerged as a promising cooling solution. This cooling method involves immersing the heat source in a virtually non-conductive coolant, delivering benefits such as simplified structure, lowering platform costs for overclocking [6], lower energy consumption, and efficient heat dissipation. ILC is further classified into two categories: phase-change immersion cooling and single-phase immersion cooling. Allied Control and 3 M use two-phase immersion cooling with 250 kW per cabinet and achieve a power usage effectiveness (PUE) of 1.02 [7]. Alibaba adopted single-phase immersion cooling, reducing energy consumption by 36 % [8]. Eiland et al. reported at an inlet temperature of 45 °C the lowest power usage effectiveness of 1.03 was achieved [9]. The superiority of ILC over conventional cooling

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means was also confirmed in the studies of Pambudi et al.[9], Qiu et al. [10], Feng et al [11], Shah et al. [12].

Current researches in ILC focuses on various aspects, including the exploration of principle, optimization of system, and design of efficient flow path within the immersion environment, among others.

Principle study based on ILC, Chhetri et al. [13] employed regression model to establish regression equations that correlate chip temperature, power consumption, and fluid flowrate under the immersion condition. Lionello et al. [14] developed a graph-based modelling approach and established an energy conservation equation to calculate the energy transport and power dissipation of components in the process of ILC. In terms of system optimisation, Huang et al [15] established a single server model to explore the effects of virtual convection, natural convection, and mixed convection methods on server cooling, and found that the average temperature of the coolant and the peak CPU temperature were reduced by 55.5 % and 40 %, respectively, and the PUE was reduced by 11.6 % when using the pump-driven mode. Matsuoka et al [16] proposed a combination of natural immersion cooling and cold plates in the cabinet and validated the feasibility of this scheme through both experimental and simulation methods. Kuncoro et al[17] used Taguchi's method to investigate the effects of fan speed, inlet flowrate, and inlet and outlet positions on the CPU temperature, and found that the cooling fan's contribution is as high as 71 %, whereas the influence of other factors is not significant. With regard to structure and flow path optimization, Cheng et al. [18] combined a heat sink with forced coolant circulation in a single-phase immersion cooling system. They observed stagnant coolant and uneven heat transfer at low flowrates, which was mitigated by increasing the flowrate. Shrigondekar et al. [19] compared the performance of PAO-6 and FC-40 coolants under seven influencing factors: inlet/outlet layouts, bypass of the fluid, suction fans, fin pitch of the heatsink, fluid flowrate, inlet temperature, and heat load. They found that due to its higher viscosity, PAO-6 exhibited a more sensitive heat transfer response on chip thermal characteristics than FC-40 when subjected to changes in external temperature. For FC-40, the best cooling effect was achieved when using a T-type inlet and outlet configuration, no bypass, and smaller heatsink fin gap sizes. Qi et al. [20] added flow distributors with long slotted and square holes to the server, reducing system pressure drop. However, at higher Reynolds numbers, the original structure had 10–15 % lower average Nusselt numbers compared to using flow distributors. Furthermore, Internal server flow baffles [21] and finned heat sinks [22,23,24] are also examples of structures that can optimize the flow field to enhance heat dissipation.

In server array immersion applications, a commonly encountered problem is uneven coolant distribution, particularly with a high number of servers in a cabinet. This uneven distribution results in poor cooling capacity for servers in areas with low flowrate, leading to an uneven temperature distribution inside the cabinet. For instance, in a study conducted by Li [25] on a cabinet with single inlet and single outlet, it was observed that the temperature difference among servers could reach up to 15 °C when the inlet mass flowrate was 10 kg/s. Sun et al. [26] demonstrated that at an inlet velocity of 0.2 m/s, the average temperature difference between chips C1 and C5 was 7 °C, with a maximum temperature difference of 10 °C. Choi [27] found that in battery pack immersion cooling, the temperature difference could be as high as 6.98 °C. However, increasing channel baffles lowered it to 6.63 °C. Furthermore, in a study by Jithin [28], the temperature uniformity of lithium-ion batteries was improved by selecting a coolant with lower viscosity and increasing the flowrate. The temperature difference between heat sources is caused by the internal coolant flow field. When extreme severe conditions occur, such as high inlet temperatures and increased server power consumption, servers with uneven flow distribution are at risk of temperature overruns, so temperature control among servers in a liquid-cooled cabinet is necessary. To enhance the distribution uniformity of flow and temperature, aside from adjusting the number of inlets and outlets, implementing baffles structures, and choosing low-viscosity coolant, this paper proposes an additional approach. It suggests the placement of a flow equalisation plate (FEP) at the bottom of the cabinet. The coolant will enter the server zone via the holes on the plate. Through careful design of the size and layout of these holes, the flow around each server can be evenly distributed. The subsequent section will delve into the specific details of this study, providing a thorough explanation.

Physical modelling and numerical methods

Physical model and performance indicators

The model of the servers and cabinet is depicted in Fig. 1(refer to TaiShan 2480 server and Scorpion 5.0 immersion liquid-cooled cabinet [29]). The maximum allowable temperature of the server is set to 85 °C. For the purpose of this paper, the models have been simplified based on the original design. The coolant enters via inlets on the cabinet's bottom sides, flowing toward the underneath of FEP model. This configuration creates a cavity between the fillers, the cabinet wall, and the FEP model. Within this cavity, the coolant undergoes rectification to ensure uniform flow before ascending to the server zone through the holes on the FEP model. The fillers inside the cabinet primarily serve to reduce the internal space, thereby conserving the coolant consumption.

Fig. 2 illustrates the FEP model. Each server is equipped with a column of holes directly beneath it. The spacing between each column of holes is the same as the spacing between the servers, measuring 121.4 mm. Each column consists of six holes, which supply coolant directly to the servers positioned above them. The spacing between rows measures 78 mm, and the FEP model has a thickness of 10 mm. The spacing between the lowest part of the server and the FEP model is 10 mm. This study investigates the effect of FEP model on the heat transfer characteristics of ILC at the server level. Since the FEP model is only employed for shunting and does not directly contact the heat source as an auxiliary heat dissipation device, the material of the FEP is not discussed.

The parameters of the server and the liquid-cooled cabinet are listed in Table 1, and for each item, the material selection is based on the corresponding materials available in the Icepak software.

To evaluate the heat transfer characteristics of ILC at the server level, the average temperature (T) of the six chips on each server is considered. Additionally, the temperature difference (ΔT) and the temperature standard deviation (T_{sd}) are used to assess the uniformity of the temperature distribution. Furthermore, the performance indicators for flow resistance is based on the pressure drop at the inlet and outlet of the cabinet. The expressions representing these indicators are as follows:

$$\Delta T = T_{\max} - T_{\min} \tag{1}$$

$$T_{sd} = \sqrt{\frac{\sum_{i=1}^{5}\left(T - T_{ave}\right)^2}{5}} \tag{2}$$

$$\Delta P = P_{in} - P_{out} \tag{3}$$

In the above expressions, ΔT is the temperature difference among the five servers, T_{max} is the maximum value of the server temperature, and T_{min} is the minimum value. T_{sd} is the standard deviation of T, and T_{ave} is the average value of the temperature of the five servers. ΔP represents the pressure drop between the inlets and outlets, P_{in} is the average inlet pressure, and P_{out} is the average outlet pressure.

Please note that in this paper, no auxiliary cooling structures such as heat sinks are employed for heat dissipation of the heat source chips. Since the coolant needs to enter the server zone through the holes on the FEP model, the hole radius, number of holes, and inlet layout configurations may have an impact on the coolant distribution, which is the focus of this paper.

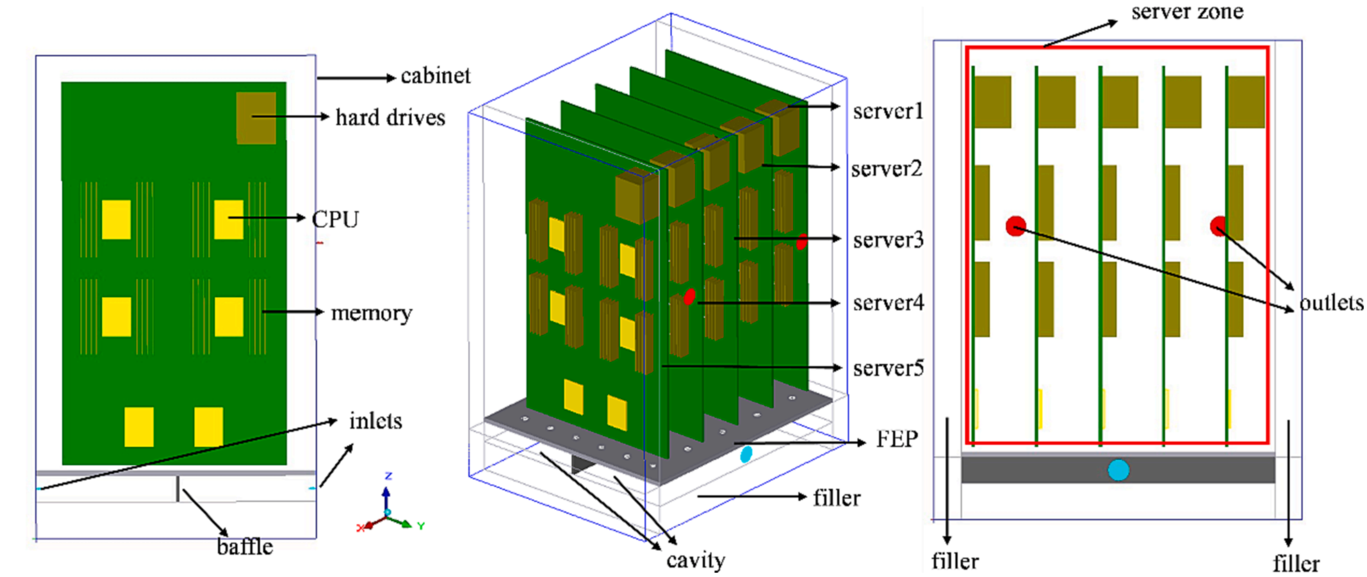

Fig. 1. The servers and cabinet model.

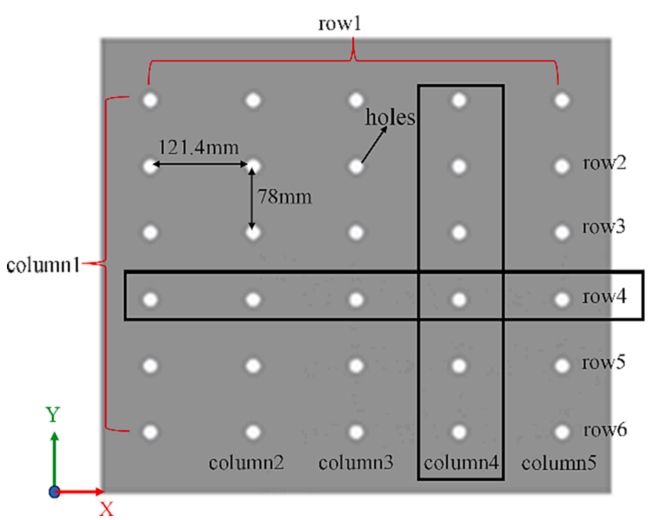

Fig. 2. The flow equalisation plate model.

Table 1
Parameters of the server and the liquid-cooled cabinet.

Item	Size(mm)	Power(W)	Quantity
CPU	5 × 55 × 75	200	6
memory	28.5 × 2.5 × 145	10	32
PCB	6.4 × 430 × 730	0	1
cabinet	707.6 × 536 × 920	–	1
FEP	10 × 536 × 602	–	1
filler block	–	–	5

Numerical assumptions and governing equations

In this study, the physical model was constructed and the calculations were performed using the Ansys Icepak software. The study was conducted based on the following assumptions: 1) The flow and heat transfer process is a steady state process. 2) Coolant physical parameters do not change with temperature. 3) Radiative heat transfer is negligible. 4) Solid surfaces apply for No Slip boundary conditions. 5) Cabinet walls are insulated.

The turbulence model selected for this study is the *Realizable k-ε* model, which offers significant advantages in predicting the diffusion of planar and circular jets. Additionally, it demonstrates improved performance in handling rotating flows, flows with strong counter-pressure gradients, flow separation, and secondary flow predicting. These characteristics make it a suitable choice for the computational requirements of this study. The first-order upwind discretization scheme is applied for momentum, temperature, turbulent kinetic energy, and turbulent dissipation rate. In addition, the *PRESTO* term is applied for the pressure term.

Eqs. (4) to (7) outline the heat transfer and flow within the cabinet. To ensure convergence and computational accuracy, the target residuals for the equations of the flow, turbulent kinetic energy, and turbulent dissipation rate are set at 1e-4, while the residuals for the energy equations are set at 1e-6.

In coolant phase:

1) Continuity equation

$$\nabla\cdot(\rho_f \vec{v}) = 0 \tag{4}$$

2) Momentum equation

$$\nabla\cdot(\rho_f \vec{v}\vec{v}) = -\nabla P + (\mu + \mu_t)\nabla^2\cdot\vec{v} + X \tag{5}$$

3) Energy equation

$$\nabla\cdot(\lambda_f \nabla T_f) = \nabla\cdot(\rho_f c_{pf} \vec{v}\cdot T_f) \tag{6}$$

In the solid phase:

1) Energy equation

$$\nabla\cdot(\lambda_s \nabla T_s) = 0 \tag{7}$$

where ρ_f, λ_f, c_{pf}, T_f represent density, thermal conductivity, specific heat capability, and temperature of the coolant, respectively, $\vec{v}$ is the velocity vector, P is the pressure, μ is the fluid viscosity, μ_t is the turbulent viscosity, X is the source term of the momentum equation, λ_s and T_s are the thermal conductivity and temperature of solid, respectively.

Thermal properties of the coolant

In this paper, FC-40 is selected as the coolant. FC-40 is a colorless, transparent, odorless perfluorinated liquid with a high boiling point and excellent thermal stability. It has found widespread use in single-phase cooling applications for semiconductor chips. The thermal physical parameters of FC-40 are listed in Table 2.

Grid and independence verification

In the numerical simulation study, the limited computational resources and the accuracy of the calculation results should be considered comprehensively. In this paper, all models were discretised by hexahedral dominant girds, and the numerical results were evaluated by varying the number of girds. Specifically, the influence of the number of girds on the accuracy of server temperature calculations was investigated under the conditions of an inlet flow velocity of 3 m/s and a hole radius of 10 mm. As observed in Fig. 3, when the number of girds reached 2,759,878, the average and maximum temperatures of the servers no longer exhibited significant changes. Therefore, the discussion was focused at this number of grids.

Model validation

To validate the model, the work conducted by Eiland et al [30] was employed. The server model can be fond in Fig. 4. As shown in Fig. 5, comparing the simulation results at inlet temperatures of 30 °C and 35 °C with the average temperature of CPU0 and CPU1 in literature [30], the maximum difference in temperature of CPUs between the simulation and experimental results is 3.3℃, and the temperature of CPUs shows a decreasing trend with increasing flowrate under both inlet temperature conditions. The variation in these two results arises from the CPU's substantial number of internal components and its complex structure. This study is based on the server level and needs to overlook this structural difference. Therefore, the model could be used to predict the heat transfer in single-phase immersion cooling.

Results and discussion

Flow distribution effect of the FEP model

To investigate the impact of flow distribution using the FEP model on server temperature uniformity, the internal flow field and temperature distribution were observed under the operational conditions of a hole radius r = 7.5 mm and an inlet flow velocity of 3 m/s. Due to the symmetrical design of the FEP model and servers, a symmetrical internal flow field was achieved within the cabinet. Various cross-sections were selected to characterize the flow pattern, including the XY cross-section along the inlet direction in the rectification cavity, as well as the YZ and XZ cross-sections passing through the holes. The inlet temperature is 25 °C for all working conditions in this paper.

As shown in Fig. 6(a) to (b), when the fluid enters the rectification cavity, it impacts the baffle positioned in the middle of the cavity and disperses outward. As the fluid flows against the cavity walls, it generates low-velocity vortex areas, characterized by decreased pressure. Fig. 6(c) to (e) illustrate the distribution of coolant flow velocity through the columns of holes, while Fig. 6(f) to (h) depict the distribution through the rows of holes. Notably, the FEP model maintains a certain velocity across each individual hole. Upon examining Fig. 6(c) to (e), it becomes evident that the holes situated at two ends of the YZ cross-section, closer to the inlet, display lower velocity. This phenomenon arises from the entry of fluid into the rectification cavity with greater

Table 2
Thermal propertites of FC-40.

ρ_f/(kg/m^3)	c_{pf} /(J/kg/K)	λ_f /(W/m/K)	β/(1/K)	μ/(Pa·s)	boiling point/℃
1870	1050	0.067	0.0012	0.004114	158

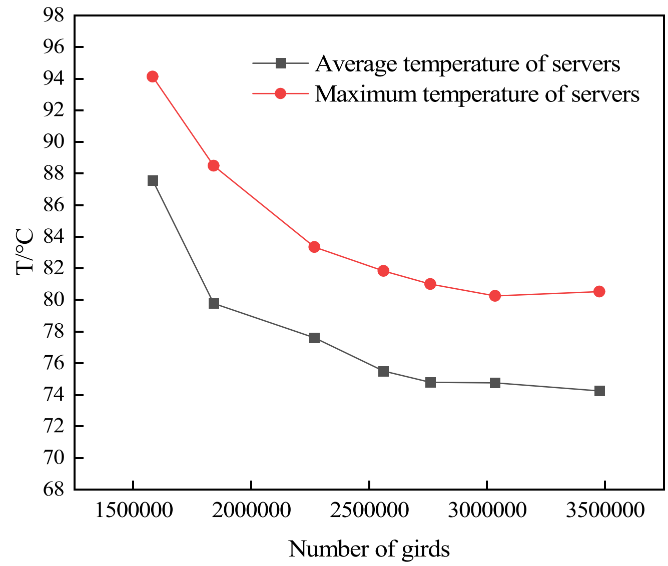

Fig. 3. Variation of the results with the number of grids.

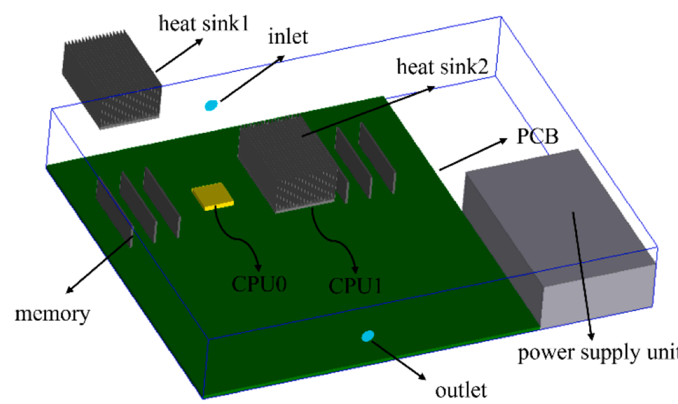

Fig. 4. The server model in the literature [30].

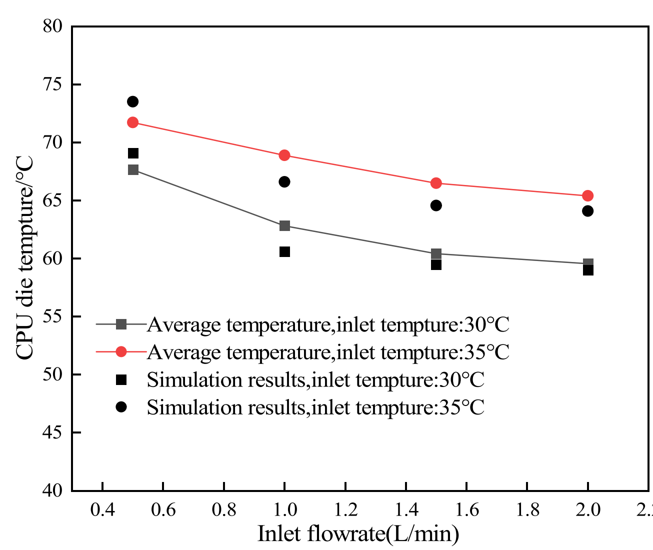

Fig. 5. Validation of the model.

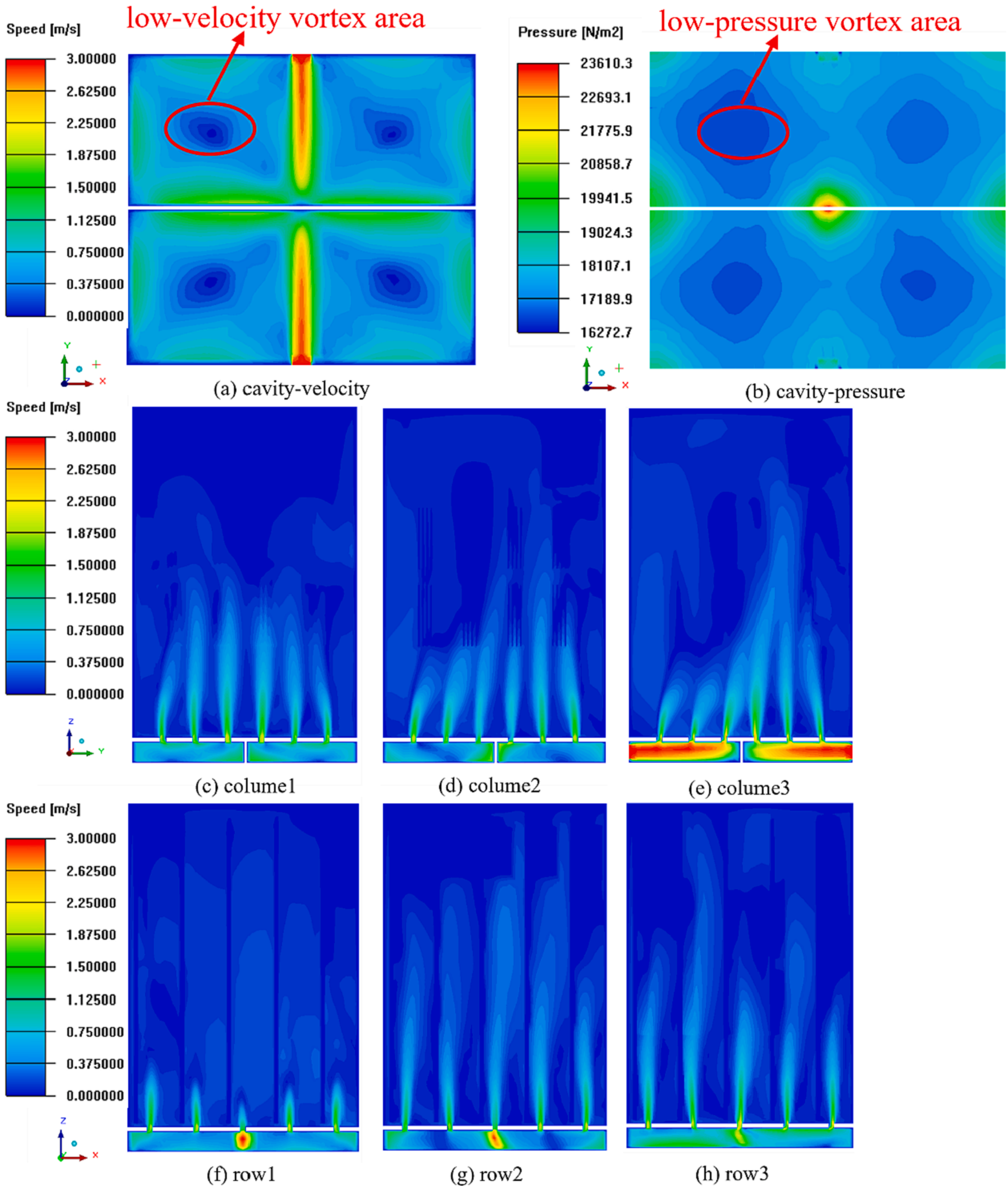

Fig. 6. Flow velocity and pressure distribution inside the cabinet. (a) Flow velocity contour inside the rectification cavity (b) Pressure contour (c) Flow velocity contour of the column1 of holes (d) column 2 (e) column 3 (f) row 1 (g) row 2 (h) row 3.

kinetic energy along the Y-axis, resulting in a lower flow velocity along the Z-axis. Moreover, as the coolant progresses and encounters the baffle obstruction, a noticeable acceleration in flow velocity occurs around the holes situated near the central region of the plate. This flow pattern is similarly apparent in Fig. 6(f) to 5(h).

Fig. 7 illustrates the average flowrate and the local maximum flowrate for each column of six holes, where each column corresponds to a server. By comparing the flowrate magnitude among the columns of holes, the flowrate around each server can be evaluated. The volumetric flowrate is provided below.

$$Q = v \times \pi \times r^2 \tag{8}$$

where v represents the flow velocity of the fluid in each hole. Fig. 7 shows that the average flowrate of each column remains stable at approximately 0.0001 m^3/s. This stability suggests a uniform flow

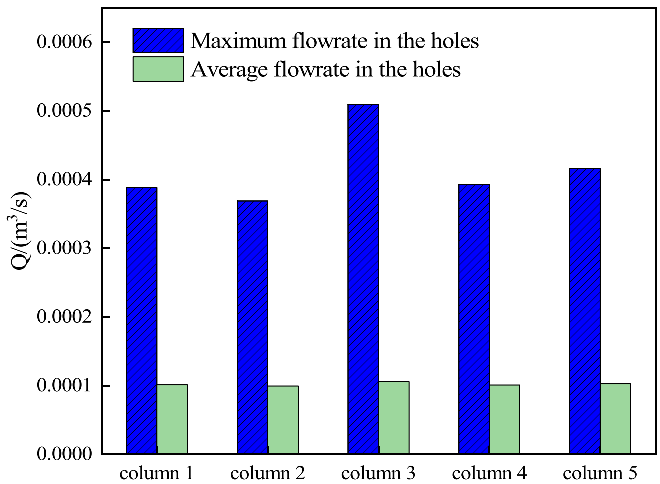

Fig. 7. The flowrate in the columns of holes.

distribution around each server. However, a discernible variation exists in the local maximum flowrates, with column 3 experiencing a 22 % to 38 % higher maximum flowrate compared to other positions. This discrepancy arises from the placement of column 3 in the path of the high-velocity inlet flow, leading to increased flowrate through its corresponding holes.

Fig. 8 shows that T of each server is stabilised at approximately 73 °C, ΔT among servers is only 1.5 °C, T_{sd} is 0.59℃, and the maximum difference in convective heat transfer coefficient (h) is 12 W/m^2/K. The gap in the maximum flowrate at the local holes, despite existing, does not disrupt the uniformity of the temperature distribution among servers and the magnitude of the convective heat transfer coefficient. The temperature of each server is indeed influenced by the flow of coolant around the chip, and during the diffusion process through the holes into the server zone, a loss of coolant flowrate occurs. As a consequence, the flowrate magnitude at the holes of the FEP model may not accurately correspond to the server temperature.

Effect of the hole radii and inlet flow velocity

In Section 3.1, the influence of temperature distribution uniformity among servers at an inlet flow velocity of 3 m/s with the FEP model and a hole radius of 7.5 mm is presented, and the flow and heat transfer inside the cabinet at different inlet velocity of r = 5 mm, r = 10 mm, and r = 12.5 mm are discussed in this section.

Fig. 9 illustrates the pressure distribution on the XY cross-section in the rectification cavity and the flow velocity in column 1 of holes under different radii, with a flow velocity of 3 m/s. The observations from Fig. 9(a) to (d) reveal a gradual decrease in cavity pressure with increasing radii, which can be attributed to the decreased flow resistance encountered by the coolant as it passes through larger holes. Correspondingly, smaller pressure lead to lower flow velocity of the coolant inside the holes, which can be seen in (e) to (h). Notably, the maximum pressure within the cavity occurs at r = 5 mm, resulting in the highest flow velocity within the holes and causing the most significant flow disturbance in the server zone.

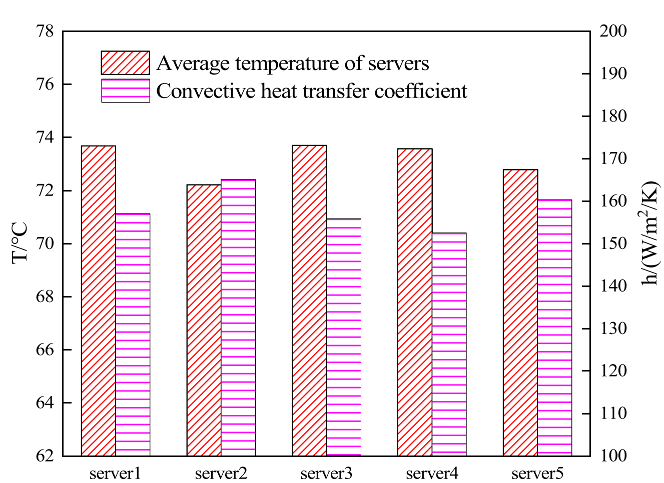

Fig. 8. Server surface temperature and convective heat transfer coefficient.

Modifying the inlet flow velocity also induces alterations in the flow velocity within the holes. As depicted in Fig. 10(a), the average velocity in all the holes on the FEP model rises significantly as the inlet flow velocity increases. The change in average velocity with inlet flow velocity is more prominent as the hole radius decreases, with the most significant response at r = 5 mm, which is attributed to the largest flow resistance of the FEP. Fig. 10(b) illustrates the variation of the maximum server temperature T_{max} with the inlet flow velocity at different radii. An increase in the inlet velocity intensifies the degree of flow disturbance around the servers, subsequently enhancing the convective heat transfer coefficient and lowering the server temperature. When modifying the velocity in the low flow velocity range, T_{max} experiences significant changes in magnitude, decreasing by approximately 3 °C for every 0.5 m/s increase. In such conditions, there is a risk of the servers exceeding the allowable temperature of 85 °C. In contrast, the change in T_{max} diminishes for flow velocity greater than 3 m/s. Moreover, as mentioned above, the use of smaller radii results in higher pressure in the cavity, leading to higher flow velocity, and therefore the server temperature continues to decrease as the radius decreases.

Fig. 11(a) to (c) depict the average flowrate Q per column of holes at flow velocities of 2 m/s, 3.5 m/s, and 5 m/s. It is evident that for r = 10 mm and r = 12.5 mm, the average flowrate in column 3, which is located in the high-velocity inlet flow direction, is greater than the flowrate in the other columns. In contrast, column 2 and column 4 are situated within the rectification cavity where low-velocity vortex flow occurs, resulting in lower flowrate. Consequently, an imbalance arises in the cabinet's flowrate. It can be seen that it is difficult to achieve effective control of the flow distribution of the coolant with larger holes. In the cases of r = 5 mm and r = 7.5 mm, nearly uniform flowrates are achieved within each column, with smaller radii resulting in higher flow resistance within the FEP model. This increased resistance makes it easier to weaken the influence of the high-velocity flow on column 3, and subsequently, helps to balance the flowrate among holes. Additionally, for r = 5 mm in Fig. 11 (d), it can be observed that the pressure drops are the largest because of the maximum flow resistance of the FEP model. Furthermore, as the flow velocity increases, the pressure loss also experiences quadratic growth. This increased pressure loss necessitates a higher pump power requirement.

The effect of radii on the uniformity of temperature distribution of the server differs in the process of flow velocity increase. According to Fig. 12, it can be observed that in the process of increasing flow velocity, when the radius is 10 mm and 12.5 mm, ΔT and T_{sd} exhibit an increasing trend in general. Notably, when r = 12.5 mm, the maximum changes in ΔT and T_{sd} are 3.3℃ and 1.0℃, respectively, which indicates a decline in the temperature distribution uniformity of servers as the velocity rises. The flow resistance in these scenarios is low and the flowrate within the column 3 is relatively high. At higher flow velocity, ΔT and T_{sd} are 4.1 °C and 1.3 °C, respectively, resulting in poor thermal management performance. In contrast, at lower flow velocity, the smaller flow resistance results in enhanced temperature uniformity, ΔT and T_{sd} are 0.8 °C and 0.3 °C, respectively. However, it's important to note that opting for larger holes will elevate the server temperature. It becomes challenging to sustain temperature distribution uniformity effectively when employing larger holes. When the radius is 5 mm and 7.5 mm, the main factor that affects the temperature uniformity of the servers is that the FEP model has larger flow resistance. Attempting to enhance

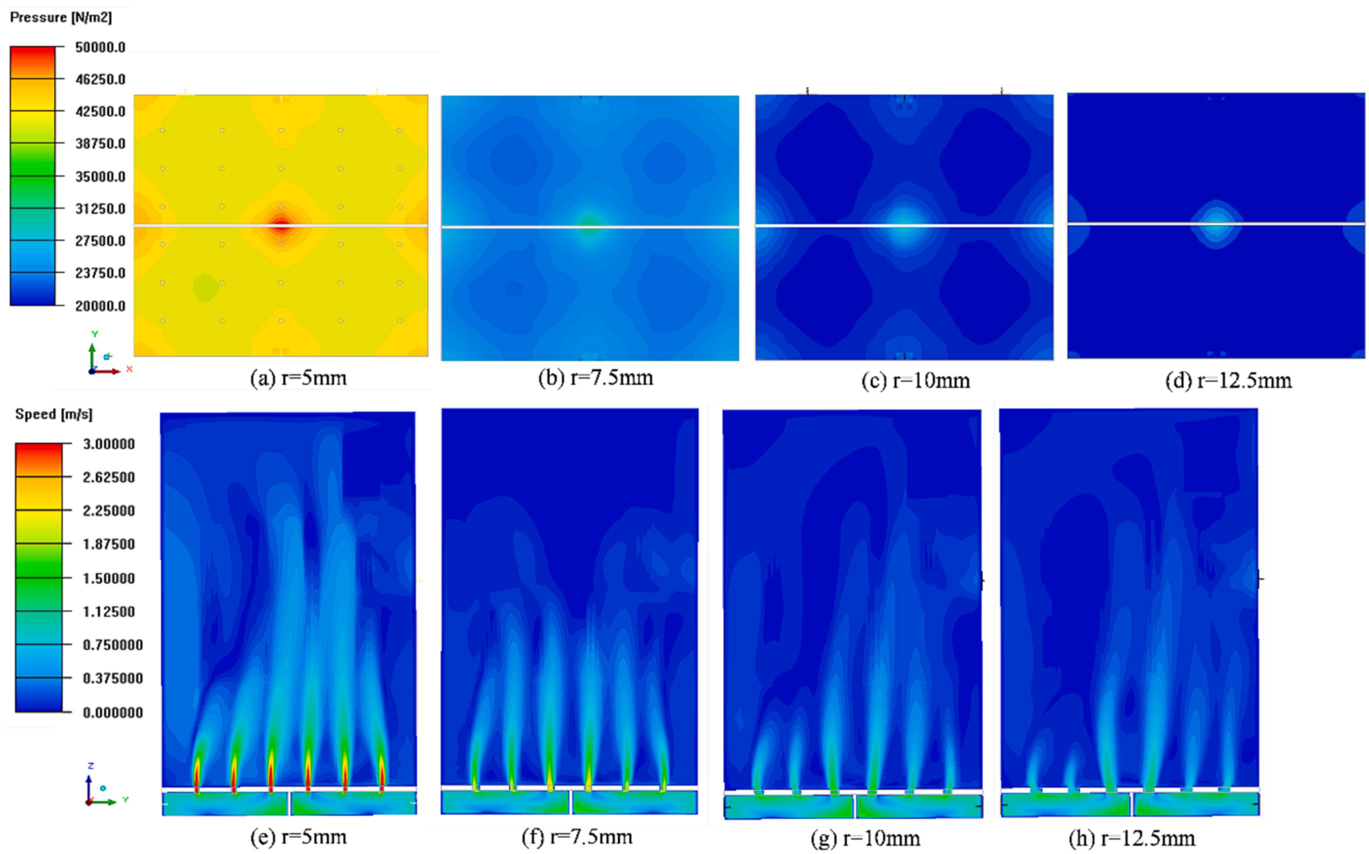

Fig. 9. Pressure and flow velocity distribution at different radius sizes. (a) Pressure contour inside the rectification cavity at r = 5 mm (b) r = 7.5 mm (c) r = 10 mm (d) r = 12.5 mm (e) Flow velocity contour in column1 of holes at r = 5 mm (f) r = 7.5 mm (g) r = 10 mm (h) r = 12.5 mm.

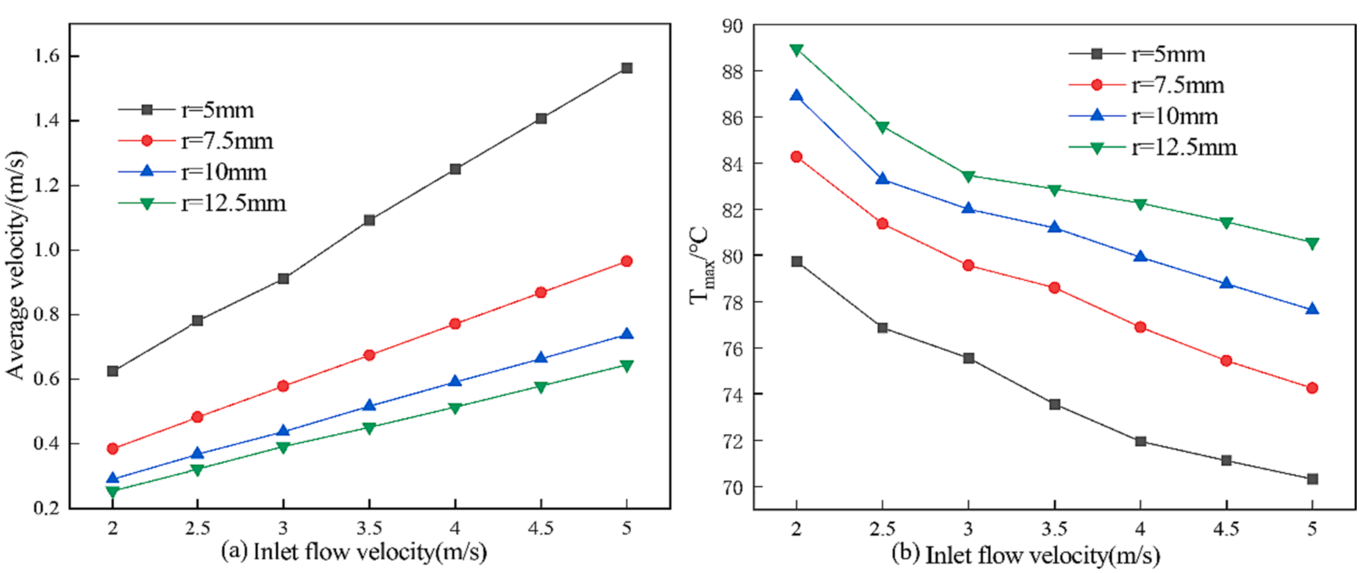

Fig. 10. (a) Average velocity in all holes variation with inlet flow velocity at different radii (b)Server temperature variation with inlet flow velocity at different radii.

performance by adjusting the flow velocity proves challenging and the change of ΔT and T_{sd} is less. When r = 7.5 mm, the flow resistance of the FEP model is lower compared to r = 5 mm, which results in a more uniform distribution of the coolant. As a result, the lowest values for ΔT and T_{sd} are achieved, at a minimum of 1.2 °C and 0.5 °C, respectively, representing the optimal temperature uniformity.

Inlet layout configurations

As mentioned earlier, when opting for larger holes, the uniformity of server temperature distribution becomes more susceptible to the flow distribution. The flow is affected by the inlet layout, so this section gives two other inlet layout schemes, as shown in Fig. 13 of the Configuration 2 and Configuration 3. The hole radius is selected as 7.5 mm and 10 mm, with an inlet flow velocity of 3 m/s.

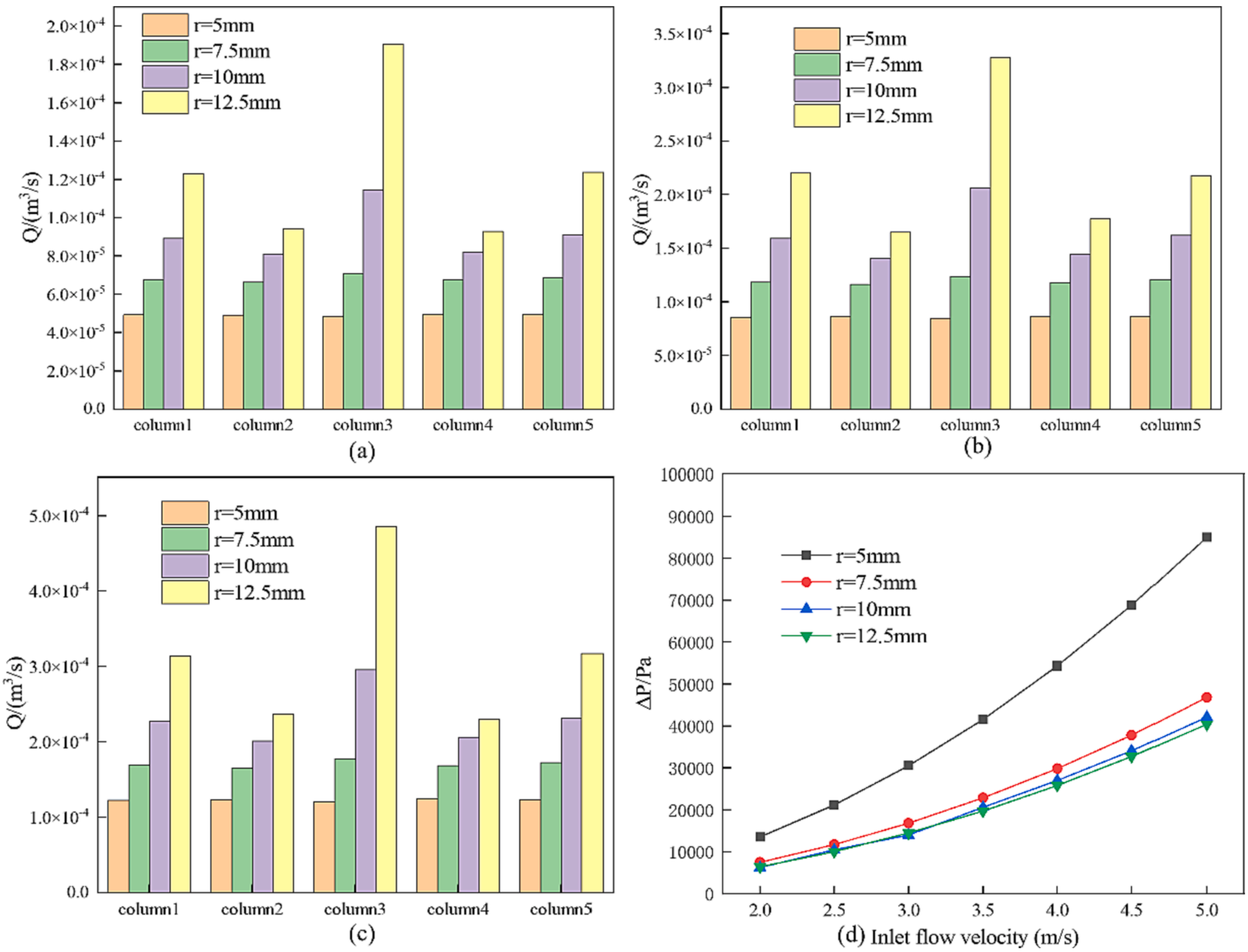

Fig. 11. Flow distribution and pressure drop variation in columns of holes with different radii and velocity (a) Flow velocity of 2 m/s (b) Flow velocity of 3.5 m/s (c) Flow velocity of 5 m/s (d) Pressure drop variation with flow velocity.

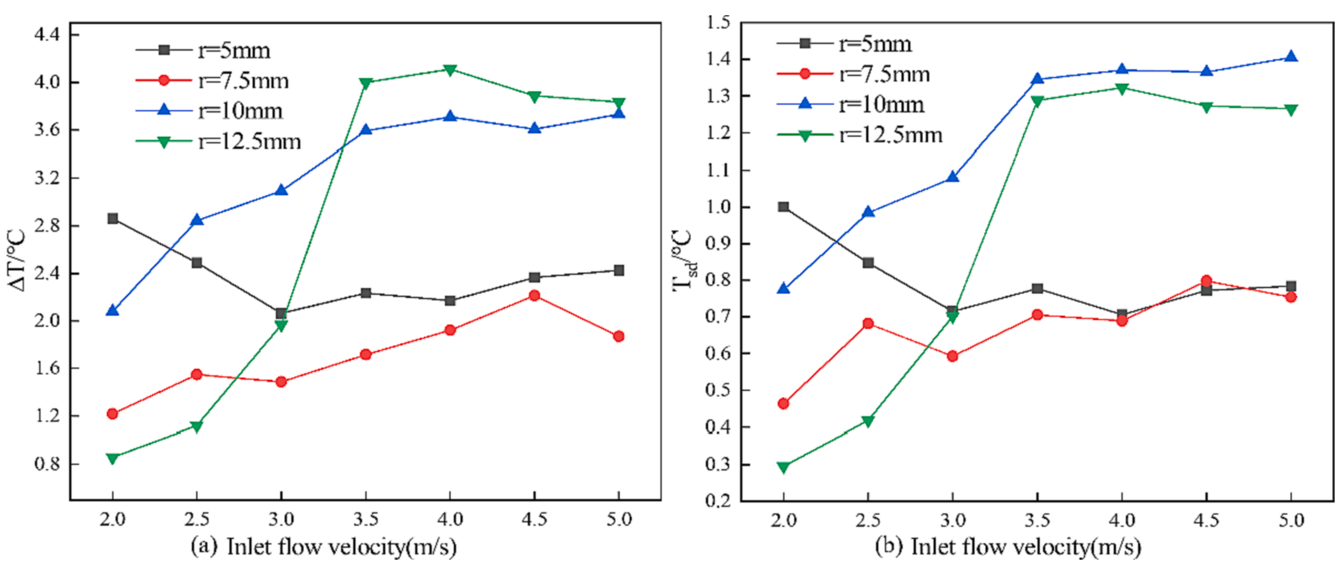

Fig. 12. Variation of server temperature uniformity with inlet flow velocity for different radius sizes (a) Variation of ΔT with flow velocity (b) Variation of T_{sd} with flow velocity.

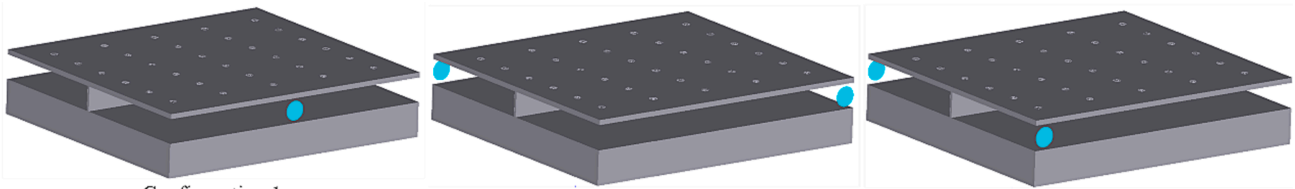

Fig. 13. Three inlet configurations.

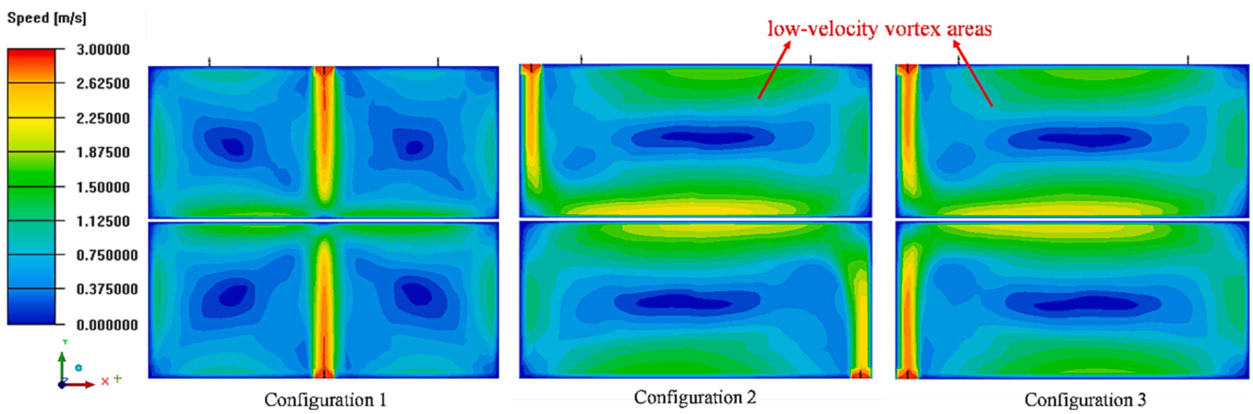

Fig. 14. Flow velocity distribution in the rectification cavity in three configurations.

Fig. 14 reveals that when inlets are positioned on the side of the cabinet (Configuration 2 and Configuration 3), the substantial low-velocity vortex areas emerges in the central region on both sides of the baffle within the rectification cavity. In Fig. 15, Configuration 2 exhibits significant flowrate differences among columns with radius of 7.5 mm and 10 mm, measuring 2.7E-05 m^3/s and 8.7E-05 m^3/s, respectively. Likewise, in Configuration 3, these differences are notable at 2.7E-05 m^3/s and 8.0E-05 m^3/s, respectively. In contrast, Configuration 1 shows relatively smaller difference, with values of 6.6E-06 m^3/s and 5.1E-05 m^3/s. Notably, the disparity in flowrate between Configuration 2 and Configuration 3 is attributed to the presence of wide low-velocity vortex areas, which results in lower flowrate in column 2, column 3, and column 4. In Configuration 1, the low-velocity vortex zone is more dispersed, leading to a more uniform coolant distribution. Furthermore, as the radius increases, the flowrate imbalance becomes more pronounced, which aligns with the conclusion drawn in the previous section.

Fig. 16 provides further confirmation of the impact of flow distribution on the temperature of servers in various configurations. When employing the inlet layout from Configuration 2 and Configuration 3, the coolant tends to accumulate on one side of the cabinet, resulting in poor temperature uniformity. This is particularly evident in Configuration 3, where, with r = 7.5 mm, ΔT reaches 2.2 °C and T_{sd} is 1.27 °C, and with r = 10 mm, ΔT increases to 6.5 °C, and T_{sd} reaches 2.4 °C. This configuration exhibits the weakest temperature control performance because of the uneven distribution of coolant, which accumulates on the same side of the cabinet, leading to inefficient flow distribution. For r = 7.5 mm, Configuration 1 demonstrates a 13 % and 34 % lower ΔT, as

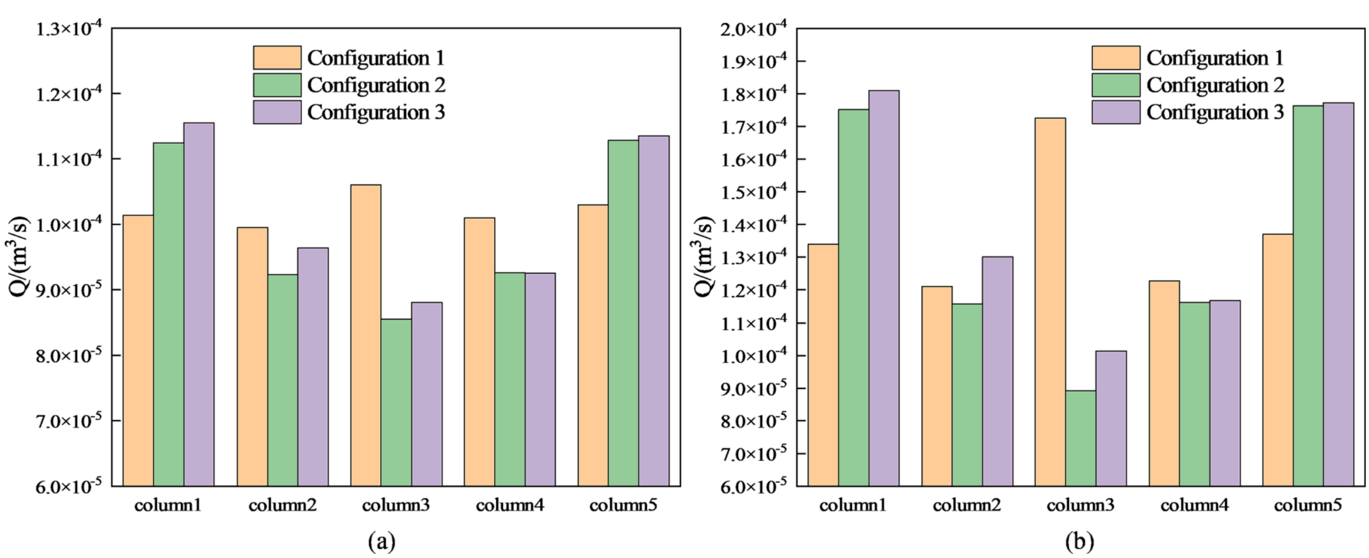

Fig. 15. Flowrate distribution between the three configurations (a) r = 7.5 mm (b) r = 10 mm.

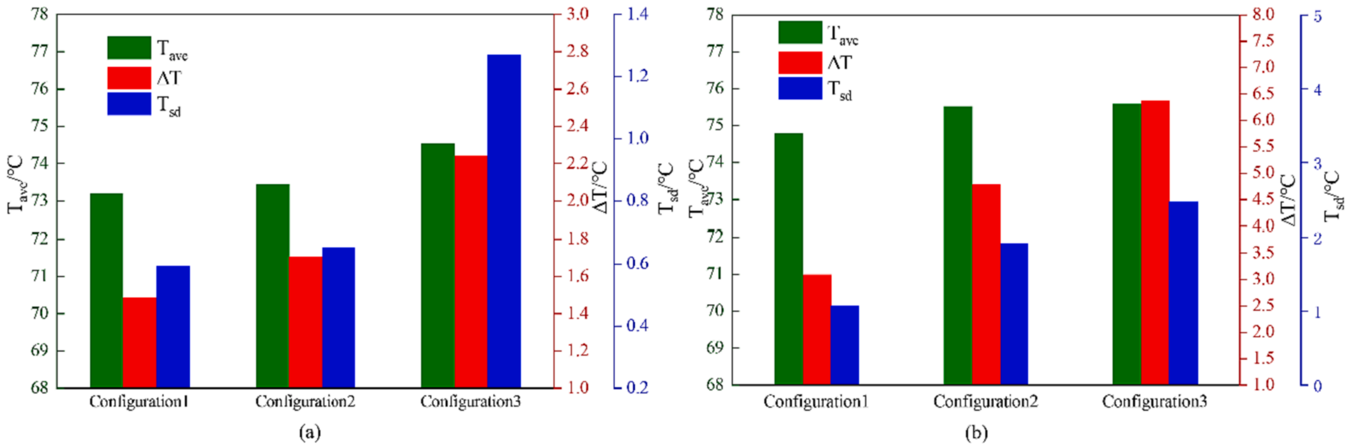

Fig. 16. T_{ave}, ΔT and T_{sd} performance in three configurations (a) r = 7.5 mm (b) r = 10 mm.

well as a 9 % and 53 % lower T_{sd}, in comparison to Configurations 2 and 3, respectively. Similarly, at r = 10 mm, Configuration 1 exhibits a 34 % and 51 % lower ΔT, along with a 44 % and 56 % lower T_{sd} than the other configurations. Furthermore, Configuration 1 yields the lowest T_{ave} and the most efficient heat dissipation. As a result, Configuration 1 excels in thermal management performance, primarily due to its superior coolant distribution.

Number of holes

The purpose of designing holes in the FEP model is to ensure a balanced temperature distribution among the servers, while also maintaining the temperature of the servers within acceptable limits. The number of holes, layout should be taken into account, as these factors collectively determine the flowrate and distribution of coolant. In the previous section, it was mentioned that each server is equipped with 6 holes underneath, resulting in a total of 30 holes. In this section, the number of holes n is reduced to 18 and 6, respectively, as shown in Fig. 17, with r = 7.5 mm and an inlet flow velocity of 3 m/s, in order to investigate the effect on heat transfer when no holes are positioned beneath the server.

As shown in Fig. 18(a) to (b), the temperature of each server decreases when the number of holes n = 18. This decrease is attributed to the increased pressure within the rectification cavity due to the reduced number of holes, consequently resulting in an increased flowrate through the holes. Since only the column 2 and column 4 of holes are removed at n = 18, the layout of the holes is still relatively uniform. Additionally, the coolant experiences higher diffusion velocity after passing through the holes, ensuring a reasonable distribution of flowrate among the servers. Nevertheless, despite this, with $\Delta T = 2.2$ °C and $T_{sd} = 0.69$ °C, the uniformity of temperature distribution experiences a somewhat milder decline compared to that at n = 30. Conversely, when n = 6, only column 3 of holes remains, leading to a low temperature of 57 °C for sever3. This configuration results in a ΔT of 14 °C and a T_{sd} of 6 °C within the cabinet, highlighting an evident lack of temperature uniformity among the servers. Notably, for n = 6, a nearly exponential rise in flow resistance within the cabinet becomes apparent due to the reduction in the number of hole, necessitating increased pump power.

In order to comprehensively evaluate the heat transfer and flow performance at n = 30, 18, and 6, a thermal performance factor ε is introduced, which is used to compare the degree of heat transfer enhancement in engineering analyses when the pump power is the same [20,31], and is defined as follows.

$$\varepsilon = (Nu/Nu_0)/(\Delta P/\Delta P_0)^{1/3} \tag{9}$$

$$Nu = \frac{h_{ave} \cdot L}{\lambda_f} \tag{10}$$

Where Nu is the Nusselt number, Nu_0 and ΔP_0 are the Nusselt number and pressure drop at n = 30, respectively, h_{ave} is the average convective heat transfer coefficient of the five servers, and L is the diameter of the inlet. The calculation shows that $\varepsilon = 1.04$ at n = 18, $\varepsilon = 0.78$ at n = 6, and n = 18 has the best overall performance.

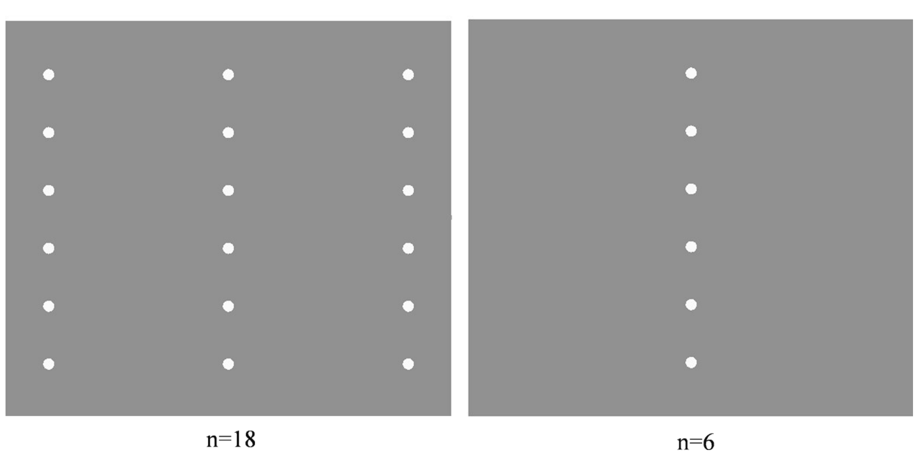

Fig. 17. Layout of holes for n = 18 and n = 6.

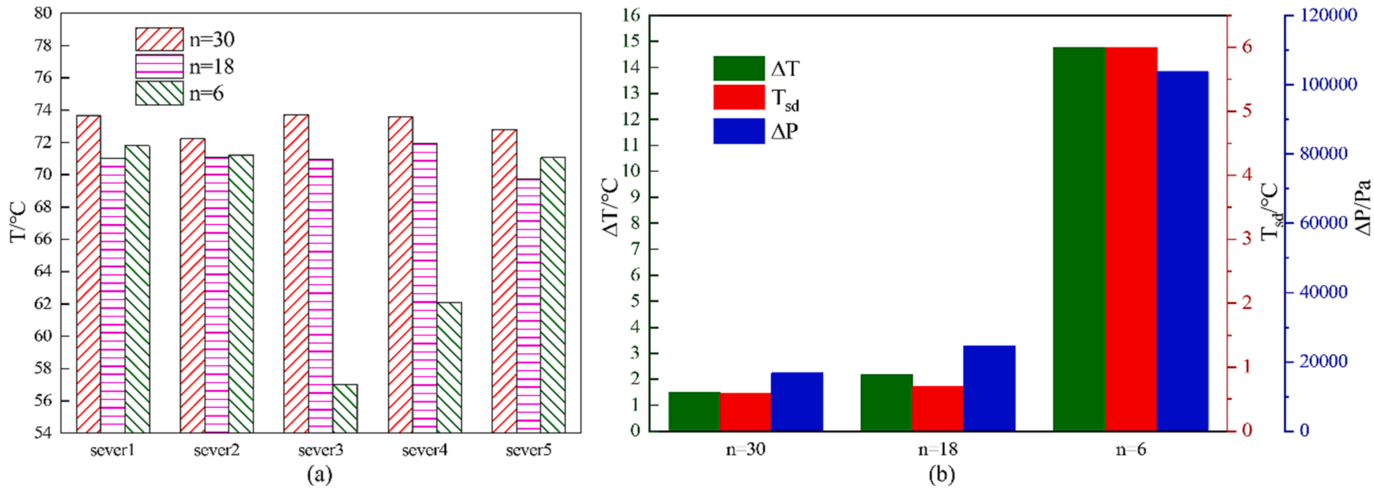

Fig. 18. (a) Temperature performance of each server with different number of holes (b) Performance of ΔT, T_{sd}, ΔP.

Conclusions

This present work studies the issue of temperature imbalance that occurs at the server level in single-phase immersion cooling. Ansys Icepak software was used to model the server, the liquid-cooled cabinet and the FEP model, and the diverting effect of the FEP to FC-40 was investigated.

1. The FEP model at the bottom of the cabinet serves to rectify the coolant before it enters the server zone. The coolant then flows through holes located at different locations and is delivered to the servers. With an inlet flow velocity of 3 m/s and a hole radius of r = 7.5 mm, it is possible to stabilize the average flowrate at 0.0001 m^3/s between the holes in each column. This configuration yields a ΔT of 1.5 °C and a T_{sd} of 0.59 °C between the servers, which effectively achieves a uniform temperature distribution and prevents localized server overheating.
2. The flow and heat transfer characteristics for various hole radii, including r = 5 mm, r = 7.5 mm, r = 10 mm, and r = 12.5 mm, are discussed respectively. As the hole size diminishes, the pressure within the rectification cavity increases, leading to higher coolant velocity through the holes. This elevation in velocity subsequently enhances the heat transfer coefficient. A larger flow resistance will make the flowrate in each column of the holes is more balanced in order to ensure the uniformity of the temperature distribution of servers. However, the increase in flowrate also results in heightened losses in flow between the inlets and outlets. Taking into account T_{ave}, ΔT, T_{sd} and P, the best performance is given by the FEP model at r = 7.5 mm.
3. When positioning the inlets at the geometric center of the rectification cavity boundary (Configuration 1), the flowrate disparity within the holes among columns is minimized. This arrangement leads to the most optimal flow distribution and temperature control, achieving $\Delta T = 1.5$ °C and $T_{sd} = 0.6$ °C at r = 7.5 mm. When arranging the inlets on the side of the cabinet (Configuration 2 and Configuration 3), the flow will be concentrated on one side. Uneven coolant distribution leads to poor thermal management performance.
4. When n = 18, the distribution of holes is uniform, the pressure inside the cavity increases, so the flowrate of the coolant increases, leading to an augmented convective heat transfer coefficient. The temperature of each server decreases, ΔT, T_{sd}, and ΔP exhibit a slight increase, with the thermal performance factor $\varepsilon = 1.04$, resulting in optimal comprehensive performance. For n = 6, the coolant flow is limited to the holes located at the base of server3. This configuration leads to highly non-uniform flow and temperature distributions. Simultaneously, a substantial flow resistance is encountered, resulting in an unusually high pressure drop of 104,000 Pa.

CRediT authorship contribution statement

Daoxu Ni: Methodology, Software, Data curation, Formal analysis, Writing – original draft, Visualization. **Fan Yu:** Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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